

sidecARM - Source File Overview

AmigaOS

`amigaos/src/Linux.c`

The main AmigaOS sidecARM launcher. It loads the Linux kernel, DTB and initramfs into shared RAM, starts the `armblk` backend, patches the device tree dynamically, and hands CPU1 over to Linux. It also contains the critical **128 MB carveout fix**.

`amigaos/src/ArmTerm-v8.c`

The AmigaOS terminal client for sidecARM. It provides direct interactive access to the Linux console through shared memory, without requiring the network stack.

`amigaos/src/armnet.device.c`

The AmigaOS SANA-II network driver. It exposes sidecARM networking to MiamiDX and exchanges Ethernet frames with Linux through shared memory.

`amigaos/src/armnet_shm.h`

Defines the shared-memory ABI used by `armnet.device`. Its structures and constants must remain compatible with the Linux-side driver.

Linux ARM64

`linux-arm64/armnet-kernel/armnet.c`

The Linux kernel network driver. It creates the `arm0` interface and exchanges Ethernet frames with the AmigaOS `armnet.device`.

`linux-arm64/armnet-kernel/armnet_shm.h`

Defines the ring buffers, slots, flags and shared-memory layout used by `armnet`.

`linux-arm64/armnet-kernel/Kconfig.snippet`

Kconfig entry required to add the ARMNET driver to the Linux kernel configuration.

`linux-arm64/armnet-kernel/Makefile.snippet`

Kernel Makefile fragment required to compile `armnet.c`.

linux-arm64/armblk-kernel/armblk.c

The Linux block-device driver for sidecARM. It exposes `/dev/armblk0`, while the actual backing image remains on the AmigaOS filesystem.

linux-arm64/armblk-kernel/armblk_shm.h

Defines the shared-memory protocol used between the Linux `armblk` driver and the AmigaOS backend.

linux-arm64/armblk-kernel/Kconfig.snippet

Kconfig entry required to enable the ARMBLK block driver.

linux-arm64/armblk-kernel/Makefile.snippet

Kernel Makefile fragment required to build `armblk.c`.

linux-arm64/armterm-kernel/armterm.c

The current Linux kernel TTY driver for ArmTerm. It replaces the old userspace `armterm-relay` and connects the Linux terminal directly to the shared-memory transport.

linux-arm64/armterm-kernel/armterm_shm.h

Defines the shared-memory ABI used by the ArmTerm kernel TTY and the AmigaOS terminal client.

linux-arm64/armterm-kernel/Kconfig.snippet

Kconfig entry required to enable the ArmTerm TTY driver.

linux-arm64/armterm-kernel/Makefile.snippet

Kernel Makefile fragment required to compile the ArmTerm driver.

linux-arm64/pinix-poc7-armblk-armnet-armterm-128M.dts

The current sidecARM device tree source. It describes the 128 MB Linux service area and the shared-memory regions used by `armblk`, `armnet` and `armterm`.

linux-arm64/initramfs-armblk-minimal/init

The minimal early userspace bootstrap. It waits for `/dev/armblk0`, mounts it as the real root filesystem, moves the pseudo-filesystems, and performs `switch_root`.

linux-arm64/initramfs-armblk-minimal/make-initramfs.sh

Build script for the minimal bootstrap initramfs used before switching to the persistent Alpine root on `/dev/armblk0`.

linux-arm64/kernel-patches/irq-bcm2836-physical-core1-poc.patch

Critical Linux IRQ patch required because sidecARM runs Linux logical CPU0 on Raspberry Pi physical core 1. It fixes the BCM2836 per-CPU interrupt routing used by timers and other local interrupts.

Emu68 changes (baseline: commit 9b4379a)

emu68-patch/armservice.c

The core Emu68 sidecARM service implementation. It manages the reserved ARM service memory and the CPU1-side control path used to start Linux alongside AmigaOS.

emu68-patch/start.c

Integrates the sidecARM service into Emu68 startup and implements the physical CPU1 Linux handoff sequence.

emu68-patch/M68k_LINE4.c

Adds the custom MOVEC interface used by AmigaOS to communicate with the ARM service. It exposes command, status, handoff offsets and the physical base address of the Linux carveout.

emu68-patch/mmu.c

Contains the Emu68 MMU changes required by the sidecARM reserved-memory layout and ARM service mapping.

emu68-patch/CMakeLists.txt

Adds the sidecARM-specific Emu68 sources to the build so the ARM service code is compiled and linked into Emu68.